

## **Quantium Virtual Internship – Task 1**

### **Executive Summary**

This analysis was conducted to understand customer purchasing behaviour within the chips category and identify the customer segments that contribute most to sales.

The transaction dataset was first cleaned by converting date formats, removing non-chip products (Salsa), and excluding an outlier customer who purchased unusually large quantities that were not representative of typical retail behaviour. Additional features such as pack size and brand name were derived from product descriptions to support further analysis.

The analysis found that the highest sales were generated by Budget Older Families, Mainstream Young Singles/Couples, and Mainstream Retirees. Older Family segments tended to purchase larger quantities of chips, while Mainstream Young Singles/Couples showed a higher willingness to pay premium prices per packet.

These findings suggest that different customer segments contribute to sales through different purchasing behaviours. Family segments drive volume through larger purchases, whereas younger mainstream customers contribute through higher-value purchases.

Based on the analysis, it is recommended that Quantum focus marketing and promotional efforts on Budget Older Families through value packs and family-oriented promotions, while targeting Mainstream Young Singles/Couples with premium products, new flavours, and targeted campaigns. Retention strategies should also be considered for Mainstream Retirees due to their strong contribution to category sales.

Overall, focusing on these key customer segments is likely to support continued growth of the chips category and improve category performance in the upcoming review period.

## Retail Strategy and Analytics – Customer Segmentation Analysis

### Objective

The objective of this analysis is to understand customer purchasing behaviour within the chips category and identify key customer segments that contribute most to sales. Insights from this analysis will support strategic recommendations for the upcoming category review.

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### 1. Import Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

plt.style.use('default')
pd.set_option('display.max_columns', None)
```

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### 2. Load Datasets

```
transactions = pd.read_excel("QVI_transaction_data.xlsx")
customers = pd.read_csv("QVI_purchase_behaviour.csv")
transactions.head()
customers.head()
```

---

### 3. Initial Data Exploration

#### Transaction Dataset Overview

```
transactions.shape
transactions.info()
transactions.describe()
transactions.isnull().sum()
```

## Customer Dataset Overview

```
customers.shape
```

```
customers.info()
```

```
customers.describe(include='all')
```

```
customers.isnull().sum()
```

## Findings

- No major missing values identified.
  - Data types appear consistent.
  - Further cleaning required on date fields and product descriptions.
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## 4. Convert Date Column

The DATE field is stored as an Excel serial number.

```
transactions["DATE"] = pd.to_datetime(  
    transactions["DATE"],  
    origin="1899-12-30",  
    unit="D"  
)  
transactions.head()
```

---

## 5. Examine Product Names

```
transactions["PROD_NAME"].head(20)
```

### Create Word Frequency Table

```
words = (  
    transactions["PROD_NAME"]  
    .str.split(expand=False)  
    .explode()  
)
```

```
words = words[
```

```
~words.str.contains(r"\d", regex=True)
]
```

```
word_counts = words.value_counts()
```

```
word_counts.head(30)
```

## Observation

The dataset contains Salsa products which do not belong to the chips category.

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## 6. Remove Salsa Products

```
transactions = transactions[
    ~transactions["PROD_NAME"]
    .str.lower()
    .str.contains("salsa")
]
transactions.shape
```

---

## 7. Detect Outliers

### Product Quantity Distribution

```
transactions["PROD_QTY"].describe()
plt.figure(figsize=(8,4))
sns.boxplot(x=transactions["PROD_QTY"])
plt.title("Boxplot of Product Quantity")
plt.show()
```

### Investigate Large Purchases

```
transactions[
    transactions["PROD_QTY"] == 200
]
```

### Examine Customer

```
transactions[
    transactions["LYLTY_CARD_NBR"] == 226000
]
```

### **Observation**

Customer 226000 purchased unusually large quantities and appears to be a commercial rather than retail customer.

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### **8. Remove Outlier Customer**

```
transactions = transactions[
    transactions["LYLTY_CARD_NBR"] != 226000
]
transactions["PROD_QTY"].describe()
```

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### **9. Transaction Trend Over Time**

```
daily_transactions = (
    transactions.groupby("DATE")
    .size()
    .reset_index(name="TRANSACTIONS")
)
plt.figure(figsize=(14,5))
plt.plot(
    daily_transactions["DATE"],
    daily_transactions["TRANSACTIONS"]
)
plt.title("Transactions Over Time")
plt.xlabel("Date")
plt.ylabel("Number of Transactions")
plt.show()
```

### **Observation**

- Transaction activity increases significantly before Christmas.
  - No transactions occur on Christmas Day, likely due to store closures.
- 

## 10. Create Pack Size Feature

```
transactions["PACK_SIZE"] = (  
    transactions["PROD_NAME"]  
    .str.extract(r"(\d+)")  
    .astype(int)  
)  
transactions["PACK_SIZE"].describe()
```

### Pack Size Distribution

```
plt.figure(figsize=(10,5))  
transactions["PACK_SIZE"].value_counts() \  
    .sort_index() \  
    .plot(kind="bar")  
  
plt.title("Transactions by Pack Size")  
plt.xlabel("Pack Size (g)")  
plt.ylabel("Transaction Count")  
plt.show()
```

### Observation

Pack sizes range from approximately 70g to 380g and appear reasonable.

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## 11. Create Brand Feature

```
transactions["BRAND"] = (  
    transactions["PROD_NAME"]  
    .str.split()  
    .str[0]  
)
```

```
transactions["BRAND"].value_counts()
```

### **Standardise Brand Names**

```
transactions["BRAND"] = (  
    transactions["BRAND"]  
    .replace(  
        "RED": "RRD"  
    )  
)  
transactions["BRAND"].value_counts()
```

---

## **12. Examine Customer Data**

### **Lifestage Distribution**

```
customers["LIFESTAGE"].value_counts()
```

### **Premium Customer Distribution**

```
customers["PREMIUM_CUSTOMER"].value_counts()
```

### **Observation**

Customer data contains no significant quality issues and is ready for analysis.

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## **13. Merge Datasets**

```
data = pd.merge(  
    transactions,  
    customers,  
    on="LYLTY_CARD_NBR",  
    how="left"  
)  
data.shape  
data.isnull().sum()
```

### **Observation**

All transaction records matched successfully with customer records.

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## 14. Create Additional Metrics

```
data["PRICE_PER_UNIT"] = (  
    data["TOT_SALES"] /  
    data["PROD_QTY"]  
)
```

---

## 15. Total Sales by Customer Segment

```
sales_segment = (  
    data.groupby(  
        ["LIFESTAGE", "PREMIUM_CUSTOMER"]  
    )["TOT_SALES"]  
    .sum()  
    .reset_index()  
)  
  
sales_segment.sort_values(  
    "TOT_SALES",  
    ascending=False  
)  
  
plt.figure(figsize=(12,6))  
sns.barplot(  
    data=sales_segment,  
    x="LIFESTAGE",  
    y="TOT_SALES",  
    hue="PREMIUM_CUSTOMER"  
)  
  
plt.xticks(rotation=45)  
plt.title("Total Sales by Customer Segment")  
plt.show()
```



## Key Finding

The highest sales are generated by:

- Budget Older Families
  - Mainstream Young Singles/Couples
  - Mainstream Retirees
- 

## 16. Number of Customers by Segment

```
customer_count = (  
    data.groupby(  
        ["LIFESTAGE", "PREMIUM_CUSTOMER"]  
    )["LYLTY_CARD_NBR"]  
    .nunique()  
    .reset_index(name="CUSTOMERS")  
)  
  
plt.figure(figsize=(12,6))  
  
sns.barplot(  
    data=customer_count,  
    x="LIFESTAGE",  
    y="CUSTOMERS",  
    hue="PREMIUM_CUSTOMER"  
)  
  
plt.xticks(rotation=45)  
plt.title("Customers by Segment")  
plt.show()
```

## Key Finding

Mainstream Young Singles/Couples and Mainstream Retirees contain the largest customer populations.

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## 17. Average Units Purchased

```
units_segment = (  
    data.groupby(  
        ["LIFESTAGE", "PREMIUM_CUSTOMER"]  
    )["PROD_QTY"]  
    .mean()  
    .reset_index()  
)  
plt.figure(figsize=(12,6))  
sns.barplot(  
    data=units_segment,  
    x="LIFESTAGE",  
    y="PROD_QTY",  
    hue="PREMIUM_CUSTOMER"  
)  
plt.xticks(rotation=45)  
plt.title("Average Units Purchased")  
plt.show()
```

### **Key Finding**

Older Families and Young Families purchase more units per transaction.

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### **18. Average Price Per Unit**

```
price_segment = (  
    data.groupby(  
        ["LIFESTAGE", "PREMIUM_CUSTOMER"]  
    )["PRICE_PER_UNIT"]  
    .mean()  
    .reset_index()  
)  
plt.figure(figsize=(12,6))
```

```
sns.barplot(  
    data=price_segment,  
    x="LIFESTAGE",  
    y="PRICE_PER_UNIT",  
    hue="PREMIUM_CUSTOMER"  
)  
plt.xticks(rotation=45)  
plt.title("Average Price Per Unit")  
plt.show()
```

### **Key Finding**

Mainstream Young Singles/Couples are willing to pay slightly higher prices per packet than other customer groups.

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## **19. Top Brands**

```
brand_sales = (  
    data.groupby("BRAND")["TOT_SALES"]  
    .sum()  
    .sort_values(ascending=False)  
)  
brand_sales.head(10)  
brand_sales.head(10).plot(  
    kind="bar",  
    figsize=(10,5)  
)  
  
plt.title("Top Brands by Sales")  
plt.show()
```

---

## **20. Preferred Pack Sizes**

```
pack_sales = (  
    data.groupby("PACK_SIZE")["TOT_SALES"]  
    .sum()  
    .sort_values(ascending=False)  
)  
pack_sales.head(10)  
pack_sales.head(10).plot(  
    kind="bar",  
    figsize=(10,5)  
)  
  
plt.title("Top Pack Sizes by Sales")  
plt.show()
```

---

## **21. Strategic Recommendations**

### **Key Insights**

1. Budget Older Families are the largest contributors to total chip sales.
2. Mainstream Young Singles/Couples represent a high-value segment due to strong customer numbers and higher spending per packet.
3. Mainstream Retirees contribute significantly through consistent purchasing behaviour.
4. Family-oriented segments purchase larger quantities of chips.
5. Mainstream Young Singles/Couples demonstrate greater willingness to pay premium prices.

### **Recommendation**

Quantum should prioritise marketing efforts toward:

- Budget Older Families through value packs and family promotions.
- Mainstream Young Singles/Couples through premium product offerings and targeted promotions.

- Mainstream Retirees through loyalty and retention initiatives. Increasing visibility of preferred brands and popular pack sizes within these segments is likely to drive additional category growth.